

```
>> dynare mcmc_counter.mod
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 23 equation(s).
Evaluating expressions...
Computing static model derivatives (order 2).
Computing static model derivatives w.r.t. parameters (order 2).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
    and 16 feedback variable(s).
Computing dynamic model derivatives (order 2).
Computing dynamic model derivatives w.r.t. parameters (order 2).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
    and 16 feedback variable(s).

Preprocessing completed.
Preprocessing time: 0h00m01s.
```

STEADY-STATE RESULTS:

q	32.8507
k	4.41224
R	1.01351
b	143.013
x	0.0406367
varphi	0.0303425
mu	0.00686924
kp	2.20776
chi	1.02177
nu	0.00432944
eta	1.28813
zeta	1.02177
N	41.8193
phi	3.41978
Ne	41.5333
Nn	0.286025
xp	6.22046
bp	-101.193
Y	3.45318
eb	0
N_obs	0
Y_obs	0
q_obs	0

EIGENVALUES:

Modulus	Real	Imaginary
0	0	0
3.882e-18	-3.882e-18	0
1.364e-17	1.364e-17	0
1.027e-14	-1.027e-14	0
4.029e-09	5.157e-15	4.029e-09

4.029e-09	5.157e-15	-4.029e-09
0.1469	0.1469	0
0.9804	0.9804	0
1.018	1.018	0
1.02	1.02	0
1.022	1.022	0
1.038	1.038	0
3.365	-3.365	0
1.193e+20	-1.193e+20	0
2.084e+20	-2.084e+20	0

There are 7 eigenvalue(s) larger than 1 in modulus for 7 forward-looking variable(s)
The rank condition is verified.

===== Identification Analysis =====

Testing prior mean

The model does not solve for prior_mean (info = 4: Blanchard & Kahn conditions are not satisfied: indeterminacy.)

Try sampling up to 50 parameter sets from the prior.

Found a random draw from the priors that solves the model:

0.0066	0.0160	0.0078	-0.3117	9.1694
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Identification now continues for this draw.

Note that differences in the criteria could be due to numerical settings, numerical errors or the method used to find problematic parameter sets.

Settings:

Derivation mode for Jacobians:	Analytic using sylvester equations
Method to find problematic parameters:	Nullspace and multicorrelation

coefficients

Normalize Jacobians:	Yes
Tolerance level for rank computations:	robust
Tolerance level for selecting nonzero columns:	1e-08
Tolerance level for selecting nonzero singular values:	1e-03

REDUCED-FORM:

All parameters are identified in the Jacobian of steady state and reduced-form solution matrices (rank(Tau) is full with tol = robust).

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):

All parameters are identified in the Jacobian of steady state and minimal system (rank(Deltabar) is full with tol = robust).

SPECTRUM (QU AND TKACHENKO, 2012):

All parameters are identified in the Jacobian of mean and spectrum (rank(Gbar) is full with tol = robust).

MOMENTS (ISKREV, 2010):

All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol = robust).

==== Identification analysis completed ====

You did not declare endogenous variables after the estimation/calib_smoother command.
Initial value of the log posterior (or likelihood): -710.6175

```
-----
f at the beginning of new iteration,      710.6174914368
Predicted improvement: 1897518.426737398
lambda =      1; f =      999.0366625
lambda =    0.33333; f =      727.1348038
lambda =    0.11111; f =      62.2927661
lambda =    0.037037; f =     -80.8300792
lambda =    0.012346; f =    -196.2473897
lambda =    0.0041152; f =   -258.2035828
lambda =    0.0013717; f =   -225.3794585
lambda =    0.00045725; f =   -14.2925223
Norm of dx      19.481
-----
```

Improvement on iteration 1 = 968.821074219

```
-----
f at the beginning of new iteration,     -258.2035827827
Predicted improvement:      219.855665287
lambda =      1; f =     -258.1871965
lambda =    0.33333; f =     -258.2024992
lambda =    0.11111; f =     -258.2035813
lambda =    0.037037; f =     -266.8610226
Norm of dx      0.2097
-----
```

Improvement on iteration 2 = 8.657439823

```
-----
f at the beginning of new iteration,     -266.8610226056
Predicted improvement:      213.685366507
lambda =      1; f =     -266.7822794
lambda =    0.33333; f =     -266.8533999
lambda =    0.11111; f =     -266.8604994
lambda =    0.037037; f =     -266.8610207
lambda =    0.012346; f =     -270.7484976
lambda =    0.023866; f =     -259.6516540
lambda =    0.01607; f =     -270.4017012
Norm of dx      0.29134
-----
```

Improvement on iteration 3 = 3.887475025

```
-----
f at the beginning of new iteration,     -270.7484976305
Predicted improvement:      0.800977915
lambda =      1; f =     -271.6941193
Norm of dx      0.0019828
-----
```

Improvement on iteration 4 = 0.945621664

```
-----
f at the beginning of new iteration,     -271.6941192946
Predicted improvement:      1.005981885
lambda =      1; f =     -270.9912644
lambda =    0.33333; f =     -272.0955477
Norm of dx      0.0035258
-----
```

Improvement on iteration 5 = 0.401428419

```
-----
f at the beginning of new iteration,     -272.0955477140
Predicted improvement:      0.030134697
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lambda =          1; f =          -272.1408472
lambda =      1.9332; f =          -272.1567573
Norm of dx 0.00038486
----
Improvement on iteration 6 =          0.061209588
-----
f at the beginning of new iteration,      -272.1567573024
Predicted improvement:          0.029830307
lambda =          1; f =          -272.2150214
lambda =      1.9332; f =          -272.2668530
lambda =      3.7372; f =          -272.3602670
lambda =      7.2247; f =          -272.5164615
lambda =     13.967; f =          -272.7335891
Norm of dx 0.00060017
----
Improvement on iteration 7 =          0.576831822
-----
f at the beginning of new iteration,      -272.7335891246
Predicted improvement:          0.751369235
lambda =          1; f =          -273.9899959
lambda =      1.9332; f =          -274.5914192
Norm of dx 0.018444
----
Improvement on iteration 8 =          1.857830064
-----
f at the beginning of new iteration,      -274.5914191883
Predicted improvement:          0.519019143
lambda =          1; f =          -275.3379320
lambda =      1.9332; f =          -275.3783548
Norm of dx 0.012942
----
Improvement on iteration 9 =          0.786935636
-----
f at the beginning of new iteration,      -275.3783548238
Predicted improvement:          0.604196667
lambda =          1; f =          -275.7086932
lambda =      0.33333; f =          -275.6854372
lambda =      0.64439; f =          -275.7971520
Norm of dx 0.013772
----
Improvement on iteration 10 =          0.418797210
-----
f at the beginning of new iteration,      -275.7971520338
Predicted improvement:          0.022344406
lambda =          1; f =          -275.8162344
Norm of dx 0.0022598
----
Improvement on iteration 11 =          0.019082411
-----
f at the beginning of new iteration,      -275.8162344453
Predicted improvement:          0.002969447
lambda =          1; f =          -275.8204222
lambda =      1.9332; f =          -275.8211489
Norm of dx 0.00064311
----
Improvement on iteration 12 =          0.004914492
-----
f at the beginning of new iteration,      -275.8211489375

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Predicted improvement:      0.003106368
lambda =      1; f =      -275.8271837
lambda =      1.9332; f =      -275.8324876
lambda =      3.7372; f =      -275.8418487
lambda =      7.2247; f =      -275.8566369
lambda =     13.967; f =      -275.8730417
Norm of dx  0.0010333
----
Improvement on iteration 13 =      0.051892769
-----
f at the beginning of new iteration,      -275.8730417064
Predicted improvement:      0.054167128
lambda =      1; f =      -275.9793554
lambda =      1.9332; f =      -276.0749690
lambda =      3.7372; f =      -276.2503437
lambda =      7.2247; f =      -276.5561748
lambda =     13.967; f =      -277.0366124
lambda =     27; f =      -277.6188392
Norm of dx  0.023833
----
Improvement on iteration 14 =      1.745797480
-----
f at the beginning of new iteration,      -277.6188391867
Correct for low angle: 0.00305396
Predicted improvement:      3.121238524
lambda =      1; f =      -282.9755574
lambda =      1.9332; f =      -275.5520428
lambda =      1.3017; f =      -283.2411942
Norm of dx  0.93689
----
Improvement on iteration 15 =      5.622355029
-----
f at the beginning of new iteration,      -283.2411942159
Predicted improvement:      1.292286983
lambda =      1; f =      -282.7449673
lambda =      0.33333; f =      -283.6977167
Norm of dx  0.51276
----
Improvement on iteration 16 =      0.456522520
-----
f at the beginning of new iteration,      -283.6977167358
Predicted improvement:      0.356173332
lambda =      1; f =      -283.6171579
lambda =      0.33333; f =      -283.8620920
Norm of dx  0.047981
----
Improvement on iteration 17 =      0.164375294
-----
f at the beginning of new iteration,      -283.8620920299
Predicted improvement:      0.043354073
lambda =      1; f =      -283.9022408
Norm of dx  0.016715
----
Improvement on iteration 18 =      0.040148802
-----
f at the beginning of new iteration,      -283.9022408324
Predicted improvement:      0.001315625

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lambda =          1; f =          -283.9036413
Norm of dx  0.0011882
----
Improvement on iteration 19 =          0.001400463
-----
f at the beginning of new iteration,          -283.9036412952
Predicted improvement:          0.000210619
lambda =          1; f =          -283.9040537
lambda =      1.9332; f =          -283.9044188
lambda =      3.7372; f =          -283.9050707
lambda =      7.2247; f =          -283.9061302
lambda =     13.967; f =          -283.9074329
Norm of dx  0.0006497
----
Improvement on iteration 20 =          0.003791618
-----
f at the beginning of new iteration,          -283.9074329128
Predicted improvement:          0.005202136
lambda =          1; f =          -283.9172055
lambda =      1.9332; f =          -283.9251678
lambda =      3.7372; f =          -283.9374586
lambda =      7.2247; f =          -283.9500091
Norm of dx  0.0086399
----
Improvement on iteration 21 =          0.042576144
-----
f at the beginning of new iteration,          -283.9500090565
Predicted improvement:          0.041102896
lambda =          1; f =          -284.0290457
lambda =      1.9332; f =          -284.0971574
lambda =      3.7372; f =          -284.2140415
lambda =      7.2247; f =          -284.3885415
lambda =     13.967; f =          -284.5576503
Norm of dx  0.055748
----
Improvement on iteration 22 =          0.607641256
-----
f at the beginning of new iteration,          -284.5576503120
Correct for low angle: 0.00250167
Predicted improvement:          0.649520602
lambda =          1; f =          -285.3416223
Norm of dx  0.42367
----
Improvement on iteration 23 =          0.783971992
-----
f at the beginning of new iteration,          -285.3416223040
Correct for low angle: 0.00495377
Predicted improvement:          0.450496141
lambda =          1; f =          -285.9141887
Norm of dx  0.92822
----
Improvement on iteration 24 =          0.572566421
-----
f at the beginning of new iteration,          -285.9141887249
Correct for low angle: 0.00410905
Predicted improvement:          0.227201255
lambda =          1; f =          -286.0414500
lambda =      0.33333; f =          -286.0283184

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lambda = 0.64439; f = -286.0694196
Norm of dx 0.87255
----
Improvement on iteration 25 = 0.155230864
-----
f at the beginning of new iteration, -286.0694195889
Correct for low angle: 0.00129083
Predicted improvement: 0.632940217
lambda = 1; f = -283.9035796
lambda = 0.33333; f = -285.9496988
lambda = 0.11111; f = -286.1410569
Norm of dx 0.87029
----
Improvement on iteration 26 = 0.071637354
-----
f at the beginning of new iteration, -286.1410569430
Correct for low angle: 0.00493149
Predicted improvement: 0.102097098
lambda = 1; f = -286.2725250
Norm of dx 0.8117
----
Improvement on iteration 27 = 0.131468093
-----
f at the beginning of new iteration, -286.2725250358
Predicted improvement: 0.017674300
lambda = 1; f = -286.2938782
Norm of dx 0.40975
----
Improvement on iteration 28 = 0.021353123
-----
f at the beginning of new iteration, -286.2938781592
Correct for low angle: 0.00397032
Predicted improvement: 0.001522307
lambda = 1; f = -286.2901461
lambda = 0.33333; f = -286.2941363
lambda = 0.11111; f = -286.2941300
lambda = 0.2148; f = -286.2942156
Norm of dx 0.1208
----
Improvement on iteration 29 = 0.000337456
-----
f at the beginning of new iteration, -286.2942156147
Correct for low angle: 0.00124637
Predicted improvement: 0.003681689
lambda = 1; f = -286.2468656
lambda = 0.33333; f = -286.2903243
lambda = 0.11111; f = -286.2943281
lambda = 0.037037; f = -286.2944132
lambda = 0.071599; f = -286.2944527
Norm of dx 0.10041
----
Improvement on iteration 30 = 0.000237128
-----
f at the beginning of new iteration, -286.2944527431
Correct for low angle: 0.00236434
Predicted improvement: 0.001519936
lambda = 1; f = -286.2712496
lambda = 0.33333; f = -286.2926084

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lambda = 0.11111; f = -286.2944691
lambda = 0.037037; f = -286.2945275
Norm of dx 0.095162
----
Improvement on iteration 31 = 0.000074765
-----
f at the beginning of new iteration, -286.2945275082
Predicted improvement: 0.000648917
lambda = 1; f = -286.2952209
Norm of dx 0.092425
----
Improvement on iteration 32 = 0.000693403
-----
f at the beginning of new iteration, -286.2952209112
Predicted improvement: 0.000003623
lambda = 1; f = -286.2952252
Norm of dx 0.0058499
----
Improvement on iteration 33 = 0.000004251
-----
f at the beginning of new iteration, -286.2952251621
Correct for low angle: 0.00381165
Predicted improvement: 0.000000008
lambda = 1; f = -286.2952252
Norm of dx 0.00015967
----
Improvement on iteration 34 = 0.000000008
improvement < crit termination

```

Final value of minus the log posterior (or likelihood):-286.295225

MODE CHECK

Fval obtained by the optimization routine: -286.295225

RESULTS FROM POSTERIOR ESTIMATION

parameters

	prior mean	mode	s.d.	prior pstdev
gamma_q	-3.0000	-2.0775	2.5587	norm 5.0000
gamma_bY	0.0000	3.2708	1.0202	norm 5.0000

standard deviation of shocks

	prior mean	mode	s.d.	prior pstdev
eN	0.0100	0.0500	0.0051	invga 0.0050
eY	0.0100	0.0168	0.0016	invga 0.0050
eq	0.0100	0.0091	0.0013	invga 0.0050

Log data density [Laplace approximation] is 271.762506.

Estimation::mcmc: Multiple chains mode.

Estimation::mcmc: Searching for initial values...

Estimation::mcmc: Initial values found!

Estimation::mcmc: Write details about the MCMC... Ok!

Estimation::mcmc: Details about the MCMC are available in mcmc_counter/
metropolis\mcmc_counter_mh_history_0.mat

Estimation::mcmc: Number of mh files: 1 per block.
Estimation::mcmc: Total number of generated files: 2.
Estimation::mcmc: Total number of iterations: 125000.
Estimation::mcmc: Current acceptance ratio per chain:

Chain 1: 42.9624%
Chain 2: 43.1944%

Estimation::mcmc: Total number of MH draws per chain: 125000.
Estimation::mcmc: Total number of generated MH files: 1.
Estimation::mcmc: I'll use mh-files 1 to 1.
Estimation::mcmc: In MH-file number 1 I'll start at line 25001.
Estimation::mcmc: Finally I keep 100000 draws per chain.

MCMC Inefficiency factors per block

Parameter	Block 1	Block 2
SE_eN	41.158	41.319
SE_eY	26.776	27.918
SE_eq	30.963	26.712
gamma_q	19.732	20.190
gamma_bY	35.068	33.682

Convergence diagnostics results for chain 1.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 1.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_eN	0.054	0.006	0.450	0.900	
0.884	0.879				
SE_eY	0.017	0.002	0.003	0.551	
0.511	0.423				
SE_eq	0.009	0.001	0.000	0.406	
0.444	0.447				
gamma_q	-2.220	1.978	0.122	0.715	
0.701	0.683				
gamma_bY	3.530	0.935	0.050	0.726	
0.700	0.692				

Convergence diagnostics results for chain 2.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 2.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_eN	0.053	0.006	0.000	0.200	
0.202	0.245				
SE_eY	0.017	0.002	0.000	0.357	

0.391	0.410				
SE_eq	0.009	0.001	0.000	0.378	
0.349	0.336				
gamma_q	-2.187	1.927	0.023	0.593	
0.618	0.653				
gamma_bY	3.513	0.899	0.002	0.611	
0.630	0.668				

Univariate convergence diagnostic, Brooks and Gelman (1998):

Parameter 1... Done!

Parameter 2... Done!

Parameter 3... Done!

Parameter 4... Done!

Parameter 5... Done!

marginal density: I'm computing the posterior mean and covariance... Done!

marginal density: I'm computing the posterior log marginal density (modified harmonic mean)... Done!

ESTIMATION RESULTS

Log data density (Modified Harmonic Mean) is 271.692951.

parameters

	prior mean	post. mean	90% HPD interval		prior	pstdev
gamma_q	-3.000	-2.1939	-5.2857	1.0143	norm	5.0000
gamma_bY	0.000	3.5162	1.9637	4.8910	norm	5.0000

standard deviation of shocks

	prior mean	post. mean	90% HPD interval		prior	pstdev
eN	0.010	0.0534	0.0436	0.0627	invG	0.0050
eY	0.010	0.0172	0.0145	0.0199	invG	0.0050
eq	0.010	0.0093	0.0071	0.0113	invG	0.0050

--- Parameters Post. Std ---

gamma_q: 1.949486

gamma_bY: 0.914731

--- Shocks Post. Std ---

eN: 0.006037

eY: 0.001662

eq: 0.001336

Total computing time : 0h09m12s

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